Solution Architecture

Date	15 March 2026
Team ID	SWTID-2026-2085
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:
Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Presentation / Client Layer
[e.g., Web App (React), Mobile App (Flutter), Admin Dashboard]

↓↑

API Gateway / Load Balancer
[e.g., Nginx, AWS API Gateway, Express Server]

↓↑

Auth Service	Core Logic Service	External API Integrations
[e.g., JWT, OAuth2]	*[e.g., Data Processing, Machine Learning Models]*	*[e.g., Payment Gateway, Email Service]*

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Data / Storage Layer
[e.g., PostgreSQL Database, MongoDB, AWS S3]

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
[Presentation Layer]	*Provides a user-friendly interface for farmers to enter soil nutrient and environmental data and view crop recommendations.*	HTML5, CSS3, Bootstrap, JavaScript
[API Gateway]	*Receives user input, validates data, communicates with the Machine Learning model, and returns crop recommendations.*	*Python, Flask*
[Microservices]	Processes soil and environmental parameters using a trained model to predict the most suitable crop	*Scikit-learn, Pandas, NumPy, Pickle*
[Database]	Stores the crop recommendation dataset and the trained machine learning model used for predictions.	CSV Dataset, Pickle (.pkl)